

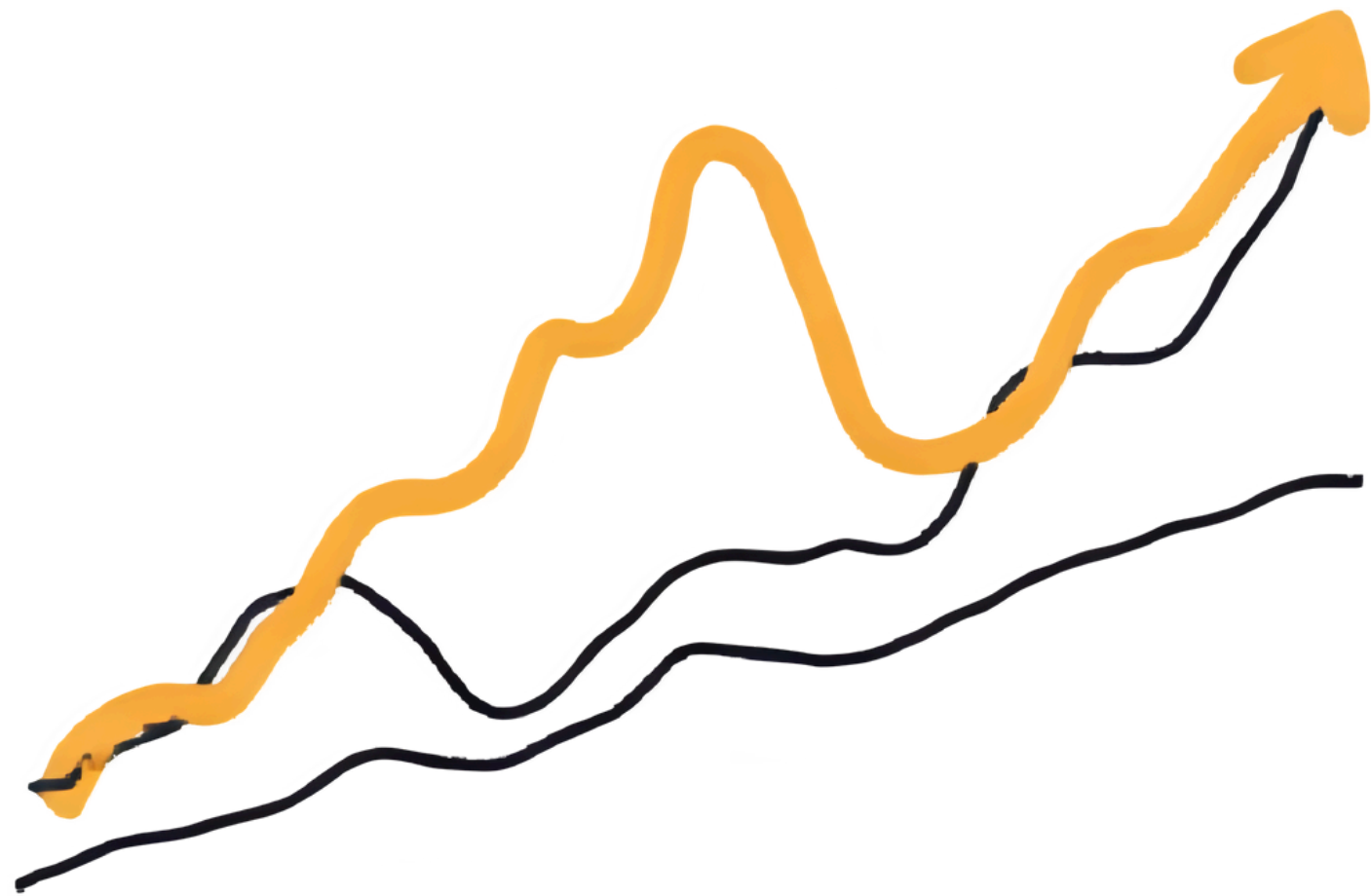
Climate Change Data Analysis

Course: Programming for Data Science and AI

Instructor: Dr. Belal Alifan

Student Name: Zain Irshadat

Project : 5 – Climate Change Indicators



Project Objectives

- Analyze real-world climate data using Python
- Identify global temperature trends over time
- Explore patterns by region and city
- Generate visual insights to support climate awareness
- Practice data cleaning and visualization techniques

Dataset Description

Source: Kaggle - Climate Change Indicators Dataset

Files Used:

- GlobalTemperatures.csv
- GlobalLandTemperaturesByCity.csv
- GlobalLandTemperaturesByCountry.csv
- GlobalLandTemperaturesByMajorCity.csv
- GlobalLandTemperaturesByState.csv

Main Columns:

- dt (date), AverageTemperature , Country, City
- Latitude, Longitude (in some files)





Libraries Used:

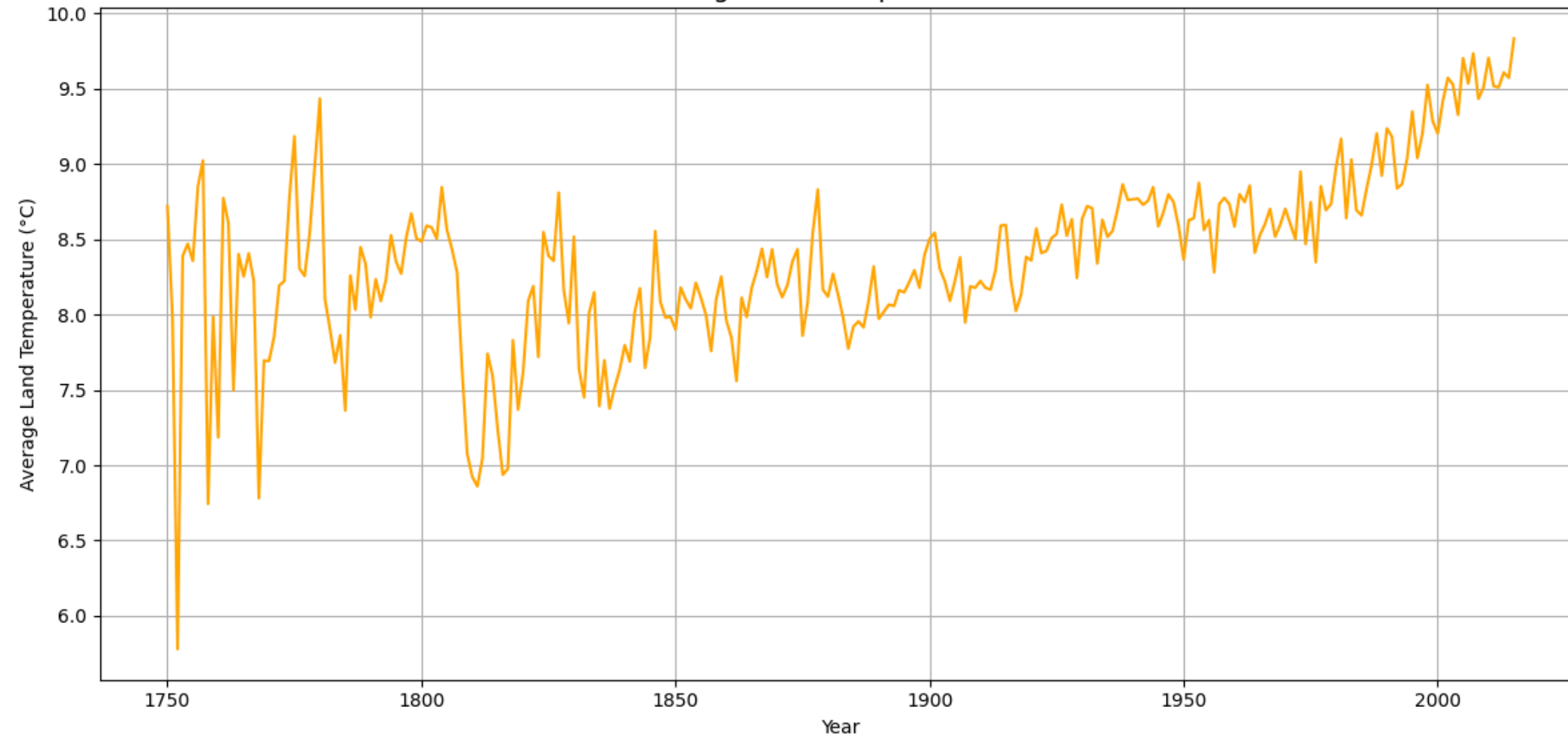
- pandas – for data cleaning and grouping
- numpy – for numeric calculations and correlations
- matplotlib – for plotting (line, bar, boxplot)
- seaborn – for styled plots and heatmaps
- math – square root for slope
- scipy.stats – linear regression
- google.colab.files – importing files
- warnings – to suppress warnings

Data Cleaning Process

- Converted date columns to datetime
- Removed missing values and duplicates
- Extracted year/month from dates
- Filtered data from 1900+ only
- Removed Antarctica from global plots
- Grouped and reset index as needed



Global Average Land Temperature Over Time

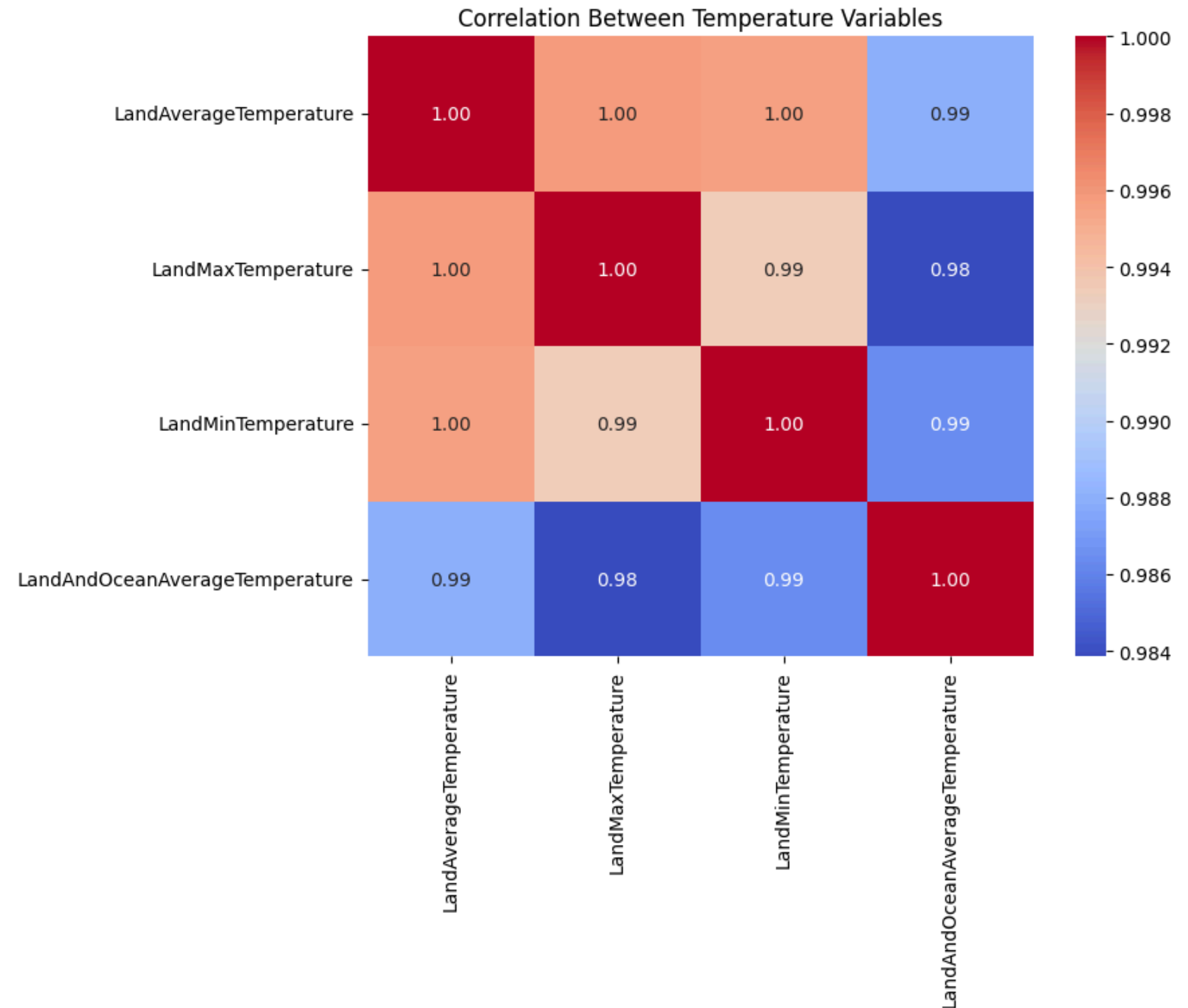


Global Average Temperature OverTime

- Grouped yearly temperature data
- Calculated average land temperature
- Clear upward trend from 1900 to 2013

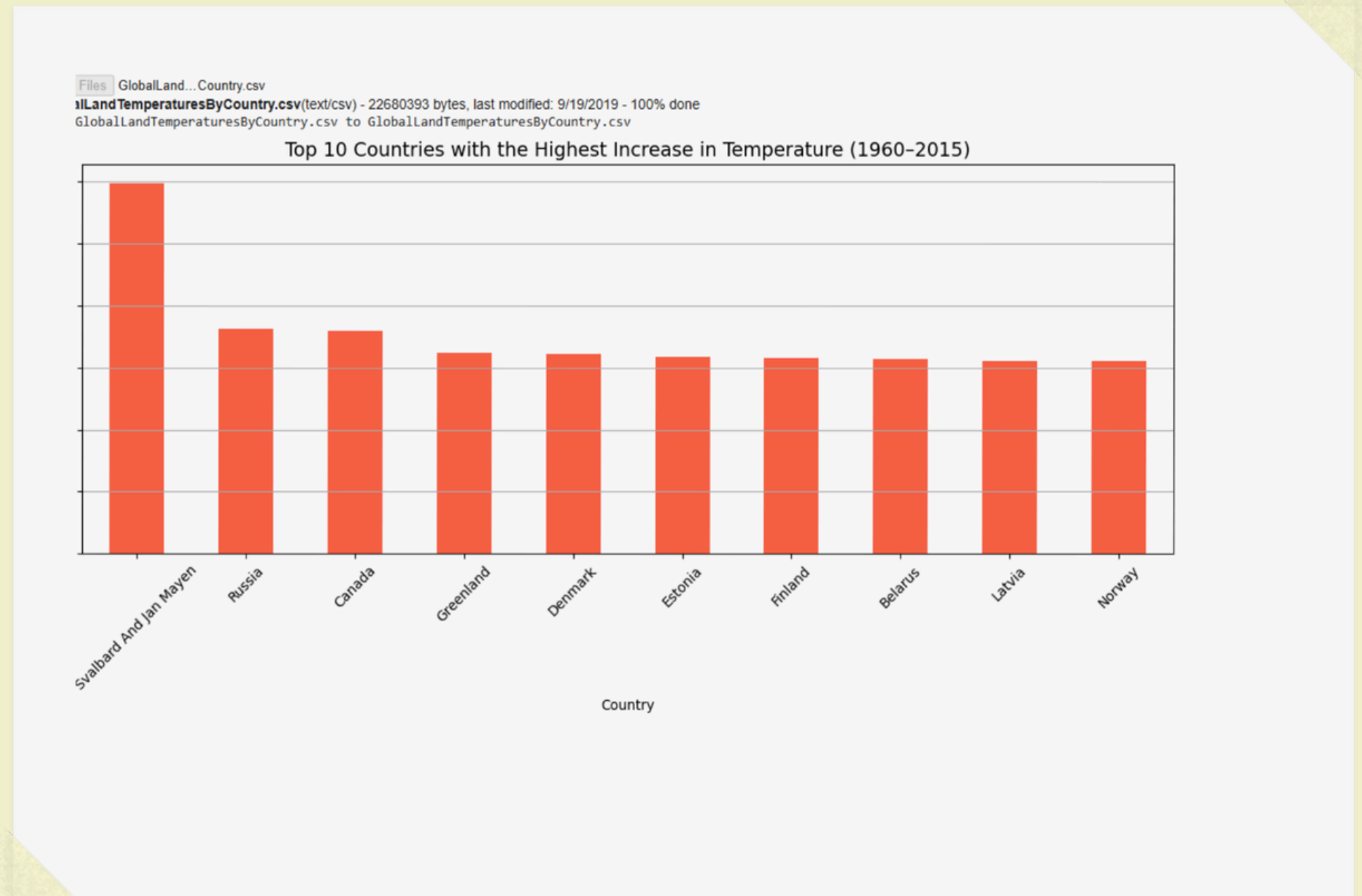
Correlation Between Temperature Variables

- Calculated Pearson correlation matrix
- Compared avg, max, min, land & ocean temps
- High positive correlation shown
- We observe that all temperature types are strongly correlated (≥ 0.98), confirming that as average temperature increases, max and min values also rise.



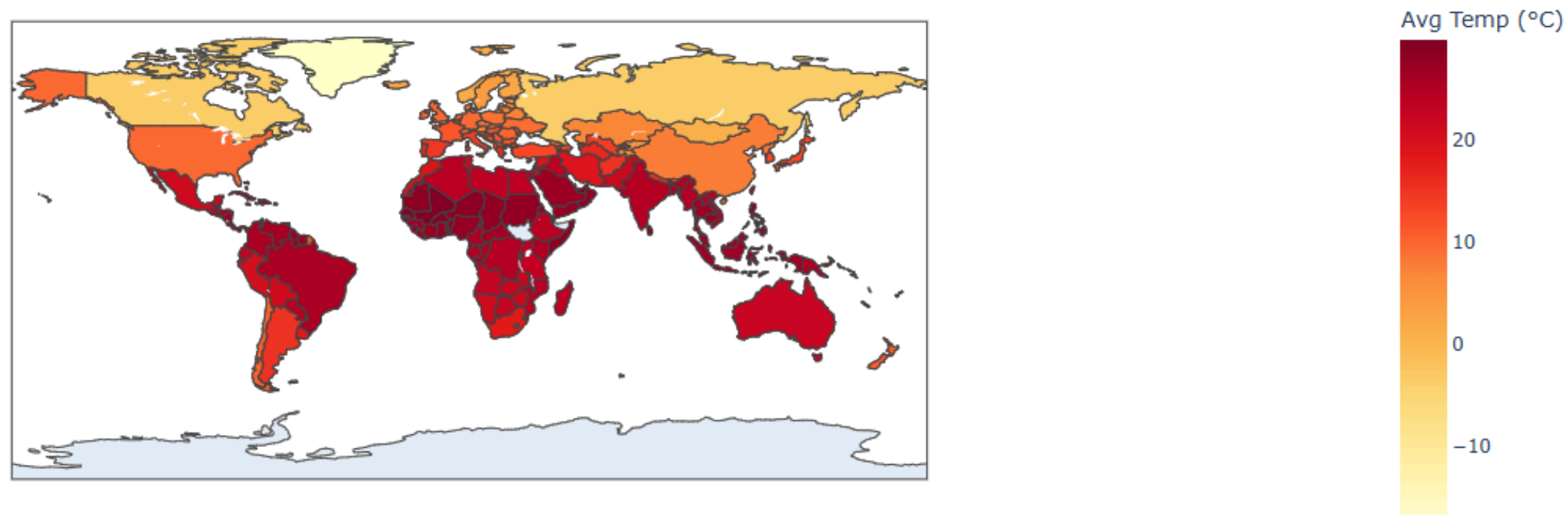
Top 10 Countries – Temperature Rise

- Applied linear regression to estimate temperature trend (slope)
- Identified top 10 countries with the steepest warming rate (1960–2015)
- Countries from the Middle East and Northern Europe dominate the list



Global Average Temperature by Country

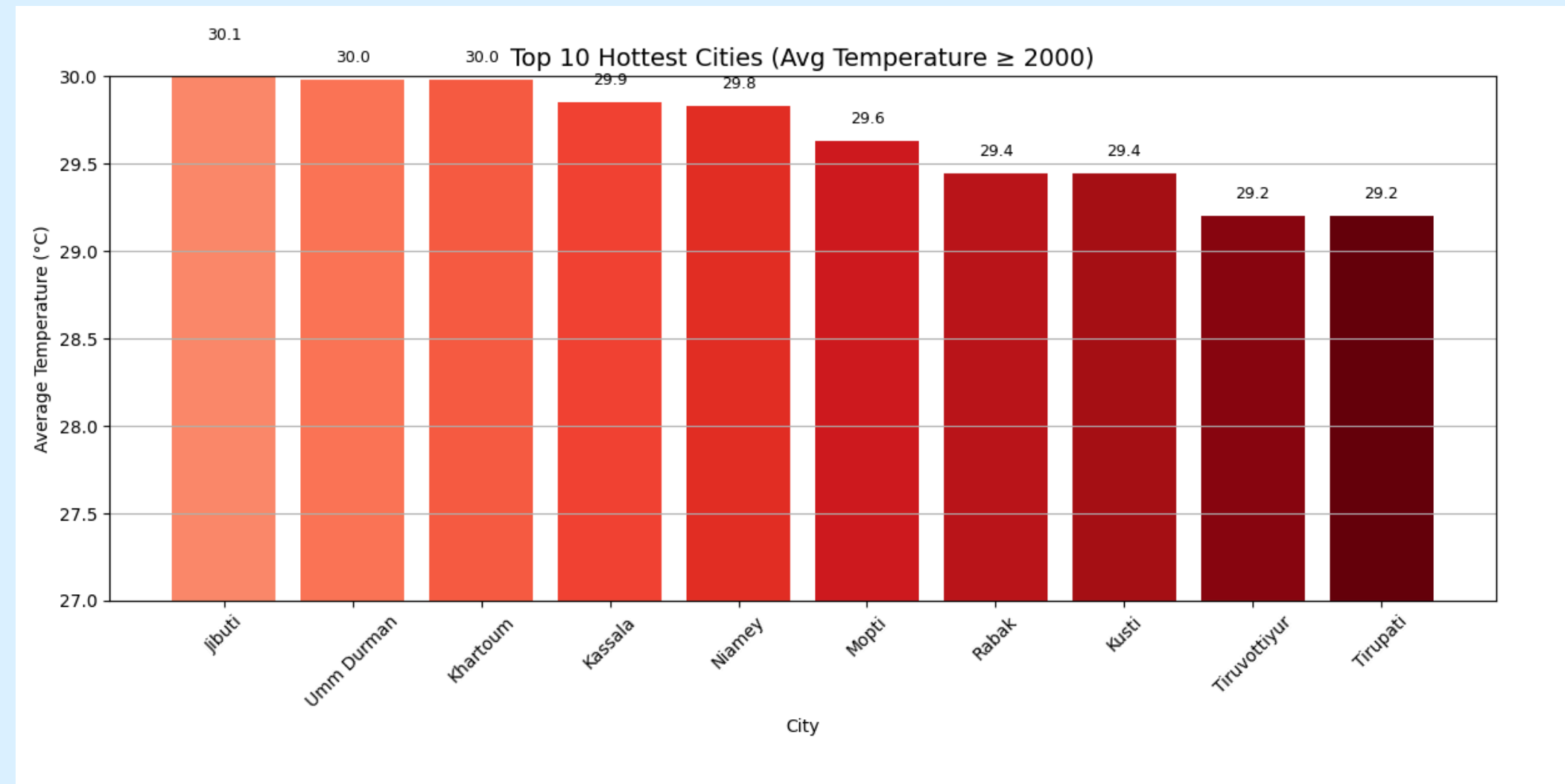
Average Temperature by Country (2000–2023)



- Calculated average temperature per country (2000–2023)
- Visualized using a world heatmap
- Highest temperatures observed in equatorial regions (e.g., Central Africa, Brazil, Southeast Asia)

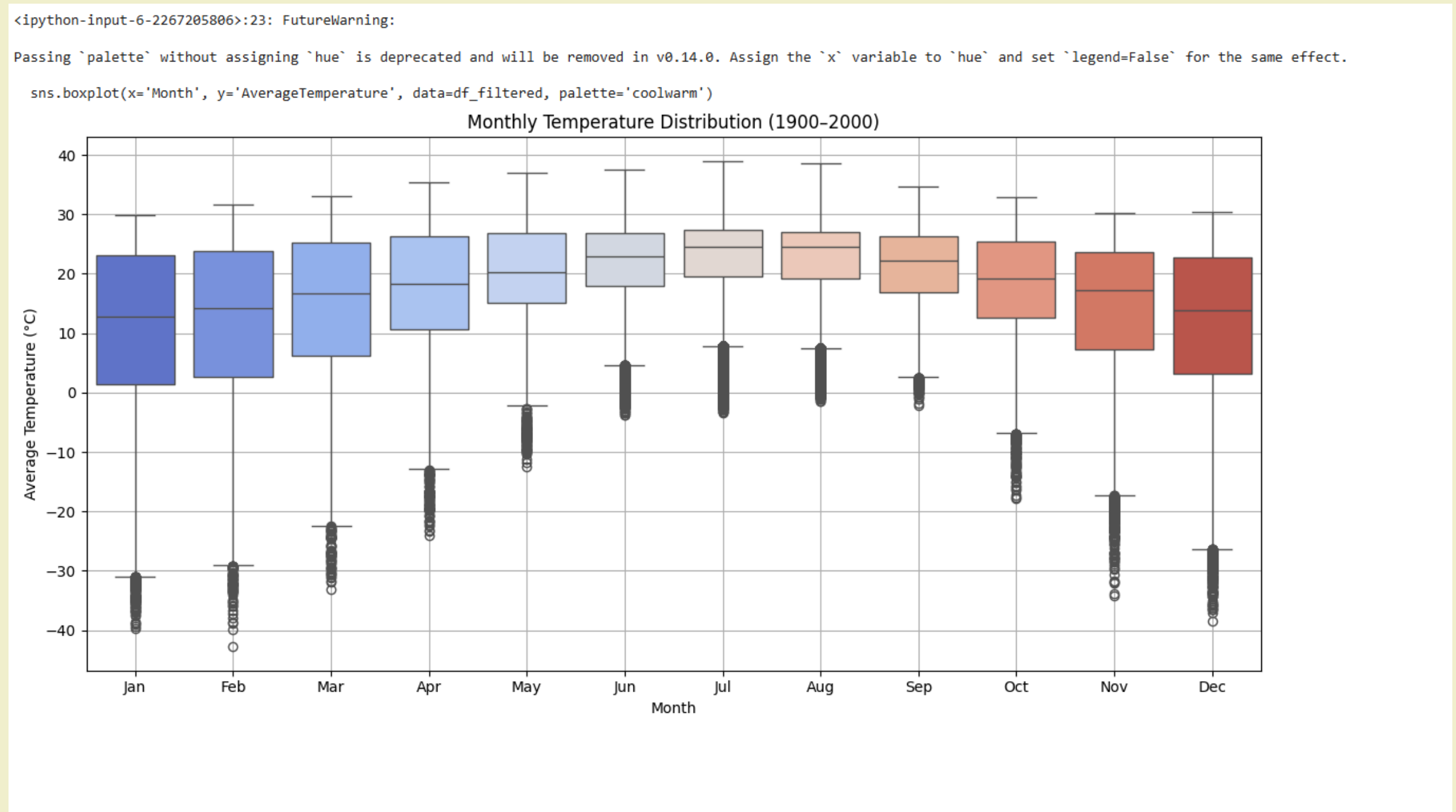
Top 10 Hottest Cities (2000–2013)

- Calculated average temperature for each city (2000–2013)
- Most of the hottest cities are located in the Gulf region
- Kuwait, Iraq, and the UAE recorded the highest urban temperatures



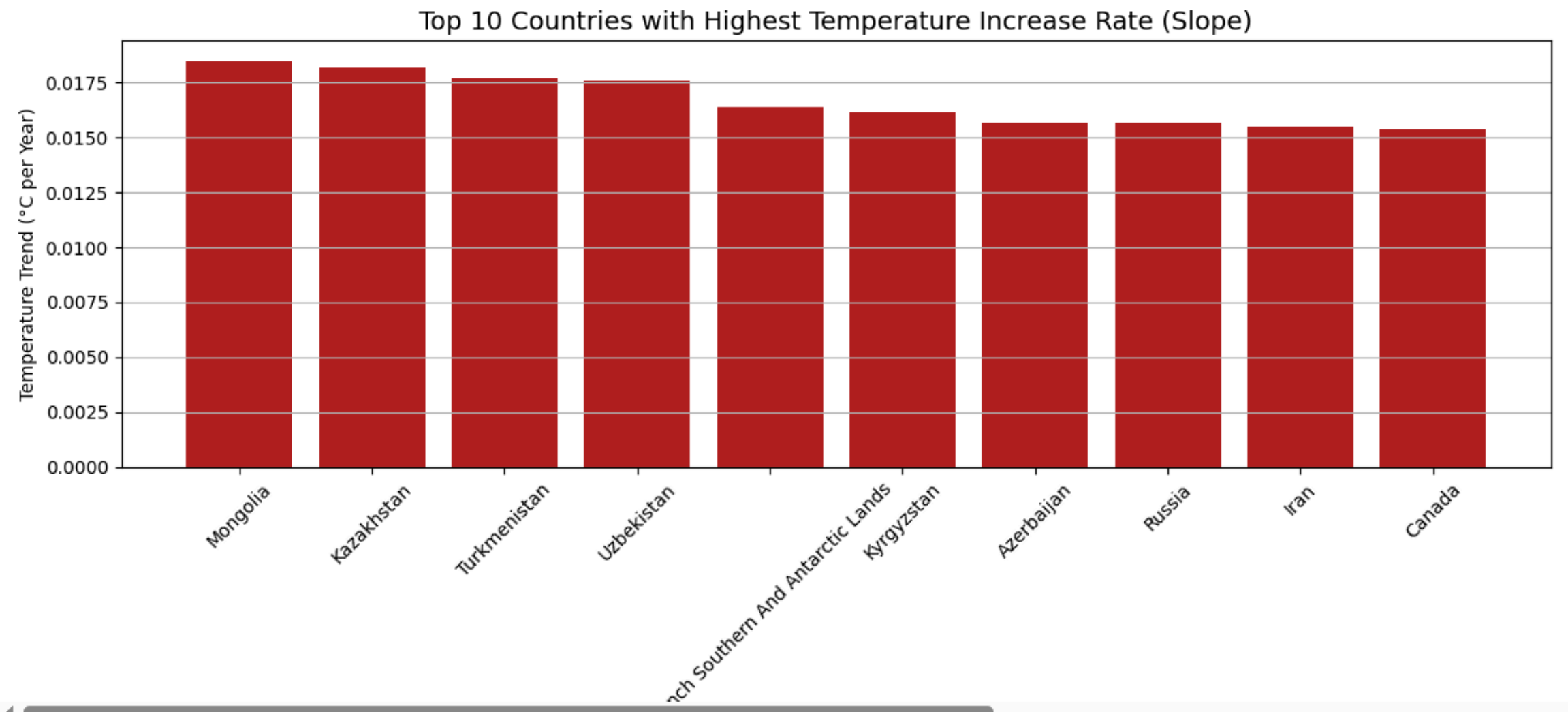
Monthly Temperature Variability (1900–2000)

- Grouped temperature data by month over 100 years
- Created a boxplot to show monthly variation
- Highest variation observed in summer months
- Winter months showed lower and more stable temperatures



Temperature Trend Slope by Country

- Applied linear regression to each country
- Calculated the slope of temperature change over time
- Most countries showed a positive slope
- Confirms a global warming pattern across regions

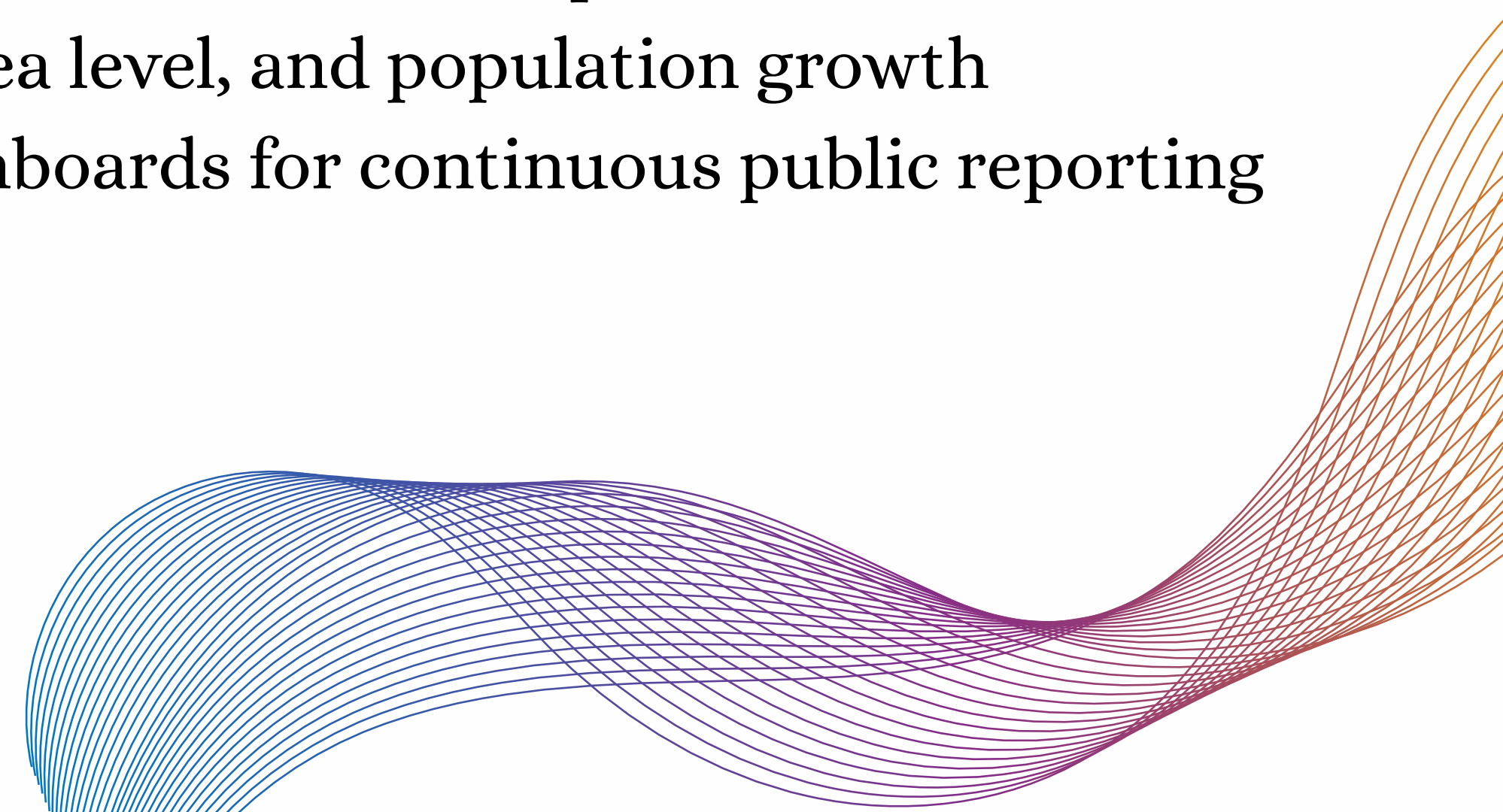


Key Insights

- Global average temperatures have steadily increased since the 1900s
- Strong positive correlations exist between all temperature variables
- Middle Eastern and African countries show the highest warming rates
- Hottest cities are concentrated in equatorial and Gulf regions
- Seasonal patterns are consistent, with summer months showing greater variation



Recommendations

- Monitor countries with the steepest warming rates using predictive models
 - Prioritize awareness campaigns in high-risk regions
 - Integrate temperature trends into national climate policies
 - Expand analysis to include CO₂, sea level, and population growth
 - Encourage the use of climate dashboards for continuous public reporting
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References

Kaggle – Climate Change Dataset

Pandas documentation

matplotlib documentation

seaborn documentation

scipy.stats (Linear Regression)

Google Colab – for running all analysis

Thank you very
much!

Made By Zain Irshadat